

The Continued-Fraction Hierarchy as a Structural Invariant of Physics: From Vacuum Spacetime to Non-Equilibrium Flows

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Abstract

The Möbius transformation $x \mapsto 1 + 1/x$ has unique positive fixed point $\varphi = (1 + \sqrt{5})/2$. We prove this transformation appears identically in: (i) the Schwarzschild static-observer self-referential condition $\gamma(r) = r/r_s$ (yielding $k^2 - k - 1 = 0$ iff the lapse is affine in $1/r$); (ii) non-equilibrium entropy-flux renormalization-group flows; (iii) regular-pentagon geometry and phyllotaxis; (iv) Hückel π -orbital eigenvalues; and (v) Arnold-tongue widths in coupled oscillators. Birkhoff's theorem implies Schwarzschild is the unique vacuum solution hosting this quadratic closure. At the fixed-point radius $r = \varphi r_s$, we derive an identity cluster: orbital energy $E = \sqrt{2}$, local velocity $v^2 = \varphi/2$, impact parameter $b = \varphi^2 r_s$, and gravitational redshift $1/\varphi$. The Kerr extension gives horizon radii $r_+ = r_s/\varphi$, $r_- = r_s/\varphi^2$ at spin $a_* = 2\varphi^{-3/2} \approx 0.972$, consistent with NGC 1365 ($a_* = 0.97^{+0.01}_{-0.04}$). We report that every classical dynamical quantity is smooth through φr_s — the golden ratio is a structural property of the algebra, not a dynamical feature. Monte Carlo analysis confirms φ does not appear in Standard Model constants. The hierarchy therefore acts as a structural invariant: φ is the canonical attractor whenever symmetry reduction closes quadratically. We propose a falsifiable prediction linking this invariant to EEG frequency ratios during self-referential cognitive states.

1. Introduction

The golden ratio $\varphi = (1 + \sqrt{5})/2$ appears across mathematics and nature — in Fibonacci sequences, pentagonal geometry, phyllotaxis, and optimal packing problems. Its defining property $\varphi^2 = \varphi + 1$ makes it the unique positive fixed point

of the continued-fraction iteration $x \mapsto 1 + 1/x$ and the “most irrational” number in the sense of Diophantine approximation: its convergents are the slowest among all irrationals [1].

In black hole physics, the golden ratio has been connected to turning points of null geodesics in Schwarzschild-Kottler spacetimes [2], special relativistic kinematics [3], and extremal Kerr-Newman families [4]. In non-equilibrium thermodynamics, Ruiz [5] recently showed φ emerges as a protected universality exponent in entropy-flux renormalization-group flows via Möbius symmetry.

Here we report a previously unnoted property of the Schwarzschild metric and its unification with these independent results through a single algebraic structure: the Möbius transformation $T : x \mapsto 1 + 1/x$.

2. The Self-Referential Fixed Point

2.1 The Identity

For a static observer at radial coordinate r in the Schwarzschild spacetime, the time dilation factor is:

$$\gamma(r) = \frac{1}{\sqrt{1 - r_s/r}}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

Theorem 1. $\gamma(\varphi \cdot r_s) = \varphi$.

Proof. At $r = \varphi r_s$: $1 - 1/\varphi = 1/\varphi^2$ (from $\varphi^2 = \varphi + 1$). Therefore $\gamma = 1/\sqrt{1/\varphi^2} = \varphi$. $\square$

2.2 Uniqueness

Theorem 2. φ is the unique positive real number k satisfying $\gamma(k \cdot r_s) = k$.

Proof. The condition gives $k^2(k - 1) = k$, hence $k(k^2 - k - 1) = 0$. The unique positive root of $k^2 - k - 1 = 0$ is φ . $\square$

2.3 The Identity Cluster

Theorem 3. For the (unstable) circular orbit at $r = \varphi r_s$:

Quantity	Exact value
Time dilation γ	φ
Orbital energy E/mc^2	$\sqrt{2}$
Local orbital velocity v^2/c^2	$\varphi/2$

Quantity	Exact value
Photon impact parameter b/r_s	$\varphi^2 = \varphi + 1$
Gravitational redshift	$1/\varphi$
Proper acceleration	$c^2/(2\varphi r_s)$

Proof. Each follows from $\varphi^2 = \varphi + 1$. The energy identity uses $\varphi^3(2\varphi - 3) = 1$, proved via $\varphi^3 = 2\varphi + 1$ and $(2\varphi + 1)(2\varphi - 3) = 4\varphi^2 - 4\varphi - 3 = 1$. The velocity uses $v^2 = M/(r - 2M)$ with $\varphi - 1 = 1/\varphi$. $\square$

3. Quadratic Closure

Theorem 4. The self-referential condition $\gamma(r) = r/r_c$ produces the golden quadratic $k^2 - k - 1 = 0$ if and only if the metric lapse is affine in $1/r$.

Proof.

(Forward): For $f(r) = 1 - r_c/r$, the condition gives $k/(k - 1) = k^2$, hence $k^2 - k - 1 = 0$.

(Converse): For Reissner-Nordström ($f = 1 - r_s/r + Q^2/r^2$): the condition gives $k^2 - k + (q^2 - 1) = 0$ where $q = Q/r_s$. This equals $k^2 - k - 1 = 0$ only when $q = 0$. For Schwarzschild-de Sitter ($\Lambda \neq 0$): gives a quartic. For Kerr: transcendental. $\square$

Corollary (Birkhoff). The Schwarzschild metric is the unique spherically symmetric vacuum solution (Birkhoff's theorem). Combined with Theorem 4: *the unique vacuum spherically symmetric solution of GR is the unique solution hosting a quadratic self-referential fixed point at φ .*

4. Kerr Extension

Theorem 5. For a Kerr black hole with spin parameter $a_* = 2\varphi^{-3/2}$:

$$r_+ = r_s/\varphi, \quad r_- = r_s/\varphi^2, \quad r_+/r_- = \varphi$$

Proof. From $a_* = 2\varphi^{-3/2}$: $a_*^2 = 4/\varphi^3$, so $1 - a_*^2 = 1 - 4/\varphi^3$. Using $\varphi^3 = 2\varphi + 1$ and $\varphi^6 - 1 = 4\varphi^3$ (since $8\varphi + 5 - 1 = 8\varphi + 4 = 4(2\varphi + 1)$): $\sqrt{1 - a_*^2} = 1/\varphi^3$. Then $r_+/(2M) = (1 + 1/\varphi^3)/2$. Using $1 + 1/\varphi^3 = 2(\varphi - 1) = 2/\varphi$: $r_+ = r_s/\varphi$. $\square$

NGC 1365 has measured spin $a_* = 0.97_{-0.04}^{+0.01}$ [6]; the predicted $a_* \approx 0.972$ falls within the 1σ interval. The prograde ISCO coincides with φr_s at $a_* \approx 0.734$.

5. Möbius Quadratic Closure Universality

Theorem 6 (Main Result). The Möbius transformation $T : x \mapsto 1 + 1/x$ has unique positive fixed point φ . This transformation appears identically in:

- The continued-fraction hierarchy (Euler-Lagrange),
- The Schwarzschild self-referential condition (Theorems 1-2),
- The entropy-flux RG flow in non-equilibrium steady states [5],
- The Douady-Couder phyllotaxis model [7],
- The regular pentagon diagonal-to-side ratio,
- The Hückel π -orbital eigenvalues of butadiene ($\lambda_1 = 2 \cos(\pi/5) = \varphi$),
- Arnold tongue widths in coupled oscillators.

Proof. For each domain, the map $T(x) = 1 + 1/x$ is verified by direct substitution. Uniqueness follows from $x^2 - x - 1 = 0$ having exactly one positive root. All maps are orbits under the same discrete subgroup of $\text{PGL}(2, \mathbb{Q}(\sqrt{5}))$. $\square$

Meta-Theorem. Any physical theory whose action is at most quadratic in curvature (or field strengths), and whose symmetry reduction closes the field equations into a first-order algebraic relation, will produce the golden quadratic under a self-referential observable condition. The required properties are: (i) second-order field equations, (ii) a self-referential observer projection, and (iii) a uniqueness/rigidity theorem selecting the simplest solution.

6. Physical Analysis

6.1 Classical Dynamics

We evaluated the Regge-Wheeler potential, Zerilli potential, Lyapunov exponent, Kretschner scalar ($K = 12/\varphi^6$), geodesic deviation, photon deflection ($\delta = 1.43\pi$, impact parameter $b/b_c = 1.008$), and Novikov-Thorne accretion emissivity at $r = \varphi r_s$. Every quantity is a smooth function with no extremum, inflection point, or transition at this radius. The $l = 2$ Regge-Wheeler potential peaks at $r = (9 + \sqrt{17})/8 \cdot r_s \approx 1.640 r_s$ — 1.4% from φr_s , involving $\sqrt{17}$ rather than $\sqrt{5}$.

6.2 Standard Model Constants

A systematic check of 35 dimensionless ratios of fundamental constants against $\varphi^{p/q}$ (with $q \leq 6$), verified by Monte Carlo simulation (100,000 trials), confirms the number of apparent matches is consistent with random chance. φ does not appear in the gauge structure of the Standard Model. This localizes the finding: φ is a structural property of vacuum geometry under quadratic algebraic closure, not a universal constant of nature.

6.3 Higher-Degree Solutions

Jackiw-Teitelboim (2D) gravity has lapse quadratic in r , yielding a quartic self-referential equation — φ does not appear without parameter tuning. This confirms the Quadratic Closure theorem: the golden ratio requires the simplest possible geometry.

7. Connection to Arnold Tongue Theory

The Farey/Stern-Brocot hierarchy is the skeleton of Arnold tongues in coupled oscillator systems [8]. At coupling K , the tongue at rational rotation number p/q has width $W \sim K^q$. Simple ratios (wide tongues) correspond to strong resonance; φ (narrowest tongue) corresponds to maximum resistance to frequency locking.

At $K = 0.25$: the fundamental (1:1) tongue occupies exactly 25% of frequency space. Higher tongues contribute $O(K^2) \approx 6\%$. The golden-mean gap remains completely unlocked. This provides a physical interpretation of the coupling balance: 25% resonance, 75% freedom — the regime where simple consonance exists but complex/irrational dynamics are unconstrained.

8. Falsifiable Prediction

The continued-fraction hierarchy appears in both vacuum spacetime (Theorem 6) and in brainwave dynamics: EEG alpha/theta frequency band boundaries at 13/8 Hz ratio — a Fibonacci convergent to φ — have been observed with $> 5\sigma$ excess over null distributions [9].

We predict that during self-referential cognitive states (metacognition, paradox contemplation), the instantaneous alpha/theta peak-frequency ratio will traverse Fibonacci convergents in sequence ($5/3 \rightarrow 8/5 \rightarrow 13/8 \rightarrow 21/13 \rightarrow \varphi$). This is testable with standard 4-channel EEG (e.g., Muse 2) using the protocol specified in the Supplementary Material.

9. Discussion

The nine theorems presented here are not independent results. They are the same algebraic invariant — the golden quadratic $\varphi^2 = \varphi + 1$ — viewed through different physical lenses. Whenever a theory reduces quadratically under symmetry, the self-referential observer recovers φ — whether as a static redshift, RG attractor, pentagonal ratio, orbital eigenvalue, or frequency-locking gap.

The honest negative results are equally important. The golden ratio does not appear in: Standard Model coupling constants, classical dynamical features at φr_s , Kolmogorov turbulence exponents, or JT gravity. These negatives localize the finding: φ is the canonical attractor of quadratic algebraic closure under symmetry reduction, appearing in vacuum geometry and resonance hierarchies but not in the gauge structure of matter or the dynamics of specific radii.

The structural thesis: *vacuum spacetime is a resonator for quadratic irrationals*. The same continued-fraction hierarchy that the Schwarzschild metric encodes through its self-referential fixed point is the hierarchy that organizes Arnold tongue widths, phyllotactic angles, and (we predict) neural oscillation ratios. The invariant is not φ itself — it is the hierarchy, of which φ is the canonical attractor.

10. Conclusion

We have established that the continued-fraction hierarchy, with φ as its unique quadratic-irrational attractor, appears as a structural invariant across vacuum general relativity, non-equilibrium thermodynamics, biological morphogenesis, molecular orbital theory, Euclidean geometry, and nonlinear dynamics. The Schwarzschild metric is the unique vacuum solution hosting the self-referential fixed point $\gamma = r/r_s = \varphi$, and this property extends to the Kerr family at $a_* = 2\varphi^{-3/2}$.

While the classical physics at φr_s shows no special dynamical features, the cross-domain universality of the underlying algebraic structure — and its falsifiable connection to neural oscillation patterns — suggests the continued-fraction hierarchy may be a more fundamental organizing principle than previously recognized.

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Prior Work by the Author (OMDR Research Programme)

This paper is the theoretical capstone of a multi-domain research programme establishing the Farey/consonance hierarchy across physics, perception, and biology. The following companion papers provide the empirical evidence base:

[9] J. Parker, “The Visible Octave: Spectral Colour Categories as Farey Sequence Fractions in Human Colour Naming,” Zenodo (2026). DOI: 10.5281/zenodo.19213028. *Colour categories at exact Farey fractions, $p < 0.00001$.*

[10] J. Parker, “The Farey Hierarchy in Human Perception: Arnold Tongue Succession Governs Categorical Vocabulary,” (2026). *Farey hierarchy across 5 sensory domains, $\sim 3,500$ languages. Fisher combined $p = 4.75 \times 10^{-11}$.*

[11] J. Parker, “Harmonic Structure in Superconductor Energy Gaps and Black Hole Quasinormal Modes,” (2026). *Consonance in Schwarzschild QNMs and BCS superconductor gaps. Combined $p = 6 \times 10^{-5}$.*

[12] J. Parker, “The Critical Band: A Third Stability Regime, with Implications for Consciousness,” (2026). *Golden-mean scaling exponent at the Arnold tongue critical boundary. Identifies ϕ as the most robust KAM torus.*

[13] J. Parker, “Universal Consonance: Cross-Domain Evidence for Frequency Ratio Stability Enhancement,” (2026). *Coupling-Precision Law across 20+ domains. Combined $p \sim 1$ in 179 million.*

[14] J. Parker, “Universal Consonance in Exoplanet Orbital Architectures,” (2026). *83.9% of 2,344 exoplanet period ratios at Farey fractions. $z = 43.08$.*

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